JIEMING ZHANG

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EDUCATION

University of Virginia **Charlottesville, VA**
B.S. Computer Engineering & B.S. Electrical Engineering, ***GPA: 3.9*** *Expected May 2029*
Relevant Coursework: Computer Systems Organization I, Electric Circuits, Digital Logic Design, CPU Architecture

TECHNICAL SKILLS

Languages: C++, Python, VHDL, Java, Assembly
EDA & Simulation: KiCad, Altium, Quartus, ModelSim
Hardware & Components: ESP32, BNO055, MicroSD SPI, Multimeters
Protocols & Tools: I2C, SPI, UART, CAN Bus, Git
Concepts: Embedded Firmware, Sensor Integration, Data Acquisition, Power Management

ENGINEERING EXPERIENCE

Virginia Motorsports (Formula SAE) **Fall 2025 - Present**
Data Acquisition (DAQ) Subteam Member | University of Virginia

- Contributing to a redesign consolidating 5 PCBs into one centralized board handling 17 analog inputs across 4 sensor types.
- Helped design schematics for the redesign and lay out the 2-layer spring 2026 prototype PCB in Altium and KiCad for sensor integration and data logging.
- Implemented an SD logging function that acquires a priority-based mutex and writes shared DAQ struct values to CSV on an SD card.

TECHNICAL PROJECTS

IoT Golf Swing Analyzer & Visualizer Prototype **Summer 2026**
Embedded Hardware, Firmware & Python Analytics | Personal Project

- Built a first-prototype ESP32 PCB (~$20 components) with 3.7V LiPo-to-3.3V power management, BNO055 I2C, and MicroSD SPI interfaces.
- Implemented firmware that logs 100 Hz quaternion and accelerometer data to CSV and a Python pipeline that applies hand-tuned drift compensation, estimates clubhead velocity, and renders 3D swing trajectories.

CPU Datapath **Spring 2026**
UVA ECE Coursework

- Built and synthesized in Quartus a 4-register × 8-bit VHDL toy CPU with ALU operations, load/store, and a conditional jump.
- Wrote testbenches and used ModelSim waveforms to find and fix opcode logic and testbench instruction errors.

Automated Physical Sink Shutoff System **Fall 2025**
Introductory Engineering Design Project | UVA

- Wrote Arduino firmware and wired an ultrasonic sensor and servo-driven arm that closes a standard sink handle.
- Tested the team prototype over about 20 trials; it shut off the sink about 80% of the time.

HACKATHON EXPERIENCE

VTHacks 14 **Sep 2026**
2-person team | Embedded Systems & Wireless Sensing

- Built ESP32 firmware that scanned 2.4 GHz Wi-Fi and Bluetooth traffic for packet counts and RSSI, storing an empty-room baseline in nonvolatile memory.
- Added a command to set a packet-count anchor and linearly rescale the density score from that anchor.